

EDUCATION

University of Wisconsin–Madison

Madison, WI

B.S. in Computer Science and Mathematics; Physics Certificate

Aug 2022 – May 2026

- **Select Coursework:** Quantum Algorithms & Error Correction, Advanced Quantum Computing, Intro to Quantum Computing, Systems Architecture for Quantum Computing, Quantum Mechanics, Cryptography, Algorithms, Analysis, Linear Algebra & Differential Equations

RESEARCH EXPERIENCE

Argonne National Laboratory

Lemont, IL

Quantum Computing Research Assistant — Distributed Quantum Computing

2025 – Present

- Advised by Dr. Rajkumar Kettimuthu.
- Enhanced ANL's SeQUeNCe library to enable distributed quantum computing simulation, implementing partitioning and scheduling algorithms to distribute logical circuits across arbitrary network topologies.
- Demonstrated 100% accuracy on a distributed three-qubit Grover's algorithm across simulated networked quantum nodes.
- Created applications for simulating quantum teleportation and remote gate operations; presented results at the 2025 Chicago Quantum Summit.

University of Wisconsin–Madison

Madison, WI

Quantum Computing Research Assistant — Benchmarking & Quantum Annealing

Summer 2023 – 2025

- Advised by Prof. Swamit Tannu (benchmarking) and Prof. Krishnan Suresh (quantum annealing).
- Developed benchmark suites with novel metrics for predicting resource expenditure and error rates on fault-tolerant quantum hardware.
- Leveraged variational autoencoders to investigate correlations between classically simulable metrics and actual quantum hardware performance.
- Adapted quantum annealing algorithms for inequality-constrained optimization with applications to finance portfolio problems; improved linear system solving efficiency by 30–40% via Box-Algorithm contraction ratio reduction.
- Applied methods to 2D-Poisson, thermal, and topology optimization problems; presented results at WCCM Vancouver 2024.

PUBLICATIONS

JCISE: S. Suresh, K. Suresh. “Computing a Sparse Approximate Inverse on Quantum Annealing Machines.”

Engineering Optimization: S. Suresh, K. Suresh. “Optimal Box Contraction for Solving Linear Systems via Simulated and Quantum Annealing.”

Textbook: S. Suresh. “A Visual Approach to Programming for Middle and High School Students.” ISBN: 979-8470081841.

PROJECTS

SurfaceScope

Python, Flask, React, TypeScript, Stim, PyMatching

Interactive Visualization Tool for Quantum Error Correction

- Developed an interactive web-based visualizer for surface-code quantum error correction, enabling step-by-step exploration of noise injection, syndrome extraction, detector error models, and minimum-weight perfect matching decoding.
- Implemented a multi-qubit lattice surgery pipeline compiling OpenQASM circuits through native gate decomposition, qubit routing with SWAP insertion, and surgery scheduling with per-slot error correction simulation.
- Built threshold estimation tooling running batch simulations across code distances and physical error rates, producing log-log plots of logical vs. physical error rates.
- Designed a responsive frontend with interactive SVG lattice rendering (pan/zoom, tooltips, manual error injection) and animated pipeline playback across six decoding stages.

Studia

Next.js, TypeScript, Convex, Vercel AI SDK

- Built a full-stack AI tutoring platform with conversational learning, checkpoint-based lesson progression, PDF context extraction for course materials, and automated performance-based review generation.

LEADERSHIP, TEACHING & AWARDS

- **Atlas Fellowship (2022):** Selected from 7,000+ applicants worldwide (114 recipients) for a \$50K STEM fellowship.
- **Printable Programming — Principal Instructor (2019 – Present):** Authored a textbook for CAD modeling, programming, and 3D-printing; taught at the Boys and Girls Club of Dane County.
- **Science Bowl — President & Captain (2020 – 2022):** Ranked 2nd nationally among individual competitors; built platforms engaging 2,000+ students.

SKILLS

Languages: Python, Java, JavaScript, C, Haskell, MATLAB

Quantum Frameworks: Qiskit, PennyLane, D-Wave Ocean, QuTiP, Stim, PyMatching

Technologies: Next.js, React, Convex, Vercel AI SDK, Flask, Turborepo, LangChain, LangGraph, Tailwind CSS

OTHER

Carnatic Violin: Professional violinist and accompanist to frontline artists at major festivals across India and the US.